

RITESH YADAV |

Software Engineer

Atlanta, GA

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WORK EXPERIENCE

Alser Lab, Georgia State University | Volunteer Researcher | Atlanta, GA | Oct 2025 – Present

- **Accelerated** Metalign bioinformatics profiling by **400% (4x speedup)** by re-architecting single-threaded routines into concurrent multi-threaded Python systems.
- **Engineered** reproducible 3-step pipeline architectures packaging Kraken, Metalign, and QIIME using Conda for deployment across laptops and high-performance computing clusters.

Georgia State University | Graduate Assistant | Atlanta, GA | May 2025 – Present

- **Optimized** digital marketing acquisition strategies across 5 university microsites by executing targeted Python data analytics on paid campaign traffic to boost conversions.
- **Designed** an interactive Power BI analytics dashboard delivering real-time program metrics to the marketing team, enhancing cross-departmental visibility for all graduate offerings.

Vyom Labs Pvt Ltd | Software Engineer | Pune, India | Mar 2021 – Jun 2023

- **Eliminated** 1,000,000+ annual manual hours across hybrid cloud infrastructure by building Python and Shell automation integrated with GitLab CI/CD, speeding builds by **200%**.
- **Architected** high-throughput **microservice REST APIs** connecting ServiceNow with BMC TSSA, processing **3,000+ weekly requests** across 50,000+ endpoints serving 50M+ users.
- **Resolved** critical distributed database bottlenecks by engineering automated maintenance routines on **PostgreSQL, Oracle DB, and SQL Server**, boosting query speed by **60%**.
- **Strengthened** platform security and regulatory compliance for 10,000+ hybrid endpoints by integrating **LDAP, RBAC**, and automated patching pipelines meeting **SOC standards**.
- **Spearheaded** technical pre-sales demonstrations and mentored 10+ junior engineers, securing contract renewals and achieving a **90%** on-time delivery rate across AWS EC2, GCP, and Azure.

Ezmata Technologies Pvt Ltd | Consultant | Pune, India | Aug 2018 – Mar 2021

- **Scaled** virtualized enterprise infrastructure to **50,000+ virtual machines** by writing custom JavaScript automation workflows within VMware vRealize Automation.
- **Secured** 3 new enterprise contracts by directing technical proof-of-concept architecture demonstrations in collaboration with cross-functional engineering teams of 20+ members.
- **Elevated** release stability and reduced production incidents by **70%** by introducing dry-run pre-flight validation routines and automated static analysis using Pester and ShellCheck.
- **Standardized** enterprise change management procedures by establishing structured documentation and staged rollout strategies across telecom, banking, and manufacturing sectors.

PROJECTS

Market News Intelligence Dashboard | Python, Ollama (Llama 3.2), Streamlit, Docker | Jan 2026 - Sep 2026

- **Developed** a real-time distributed news intelligence pipeline ingesting RSS feeds, categorizing market updates via local LLMs, and scoring mood indicators for option traders.
- **Implemented** a dynamic feedback loop utilizing few-shot prompt examples from user corrections to continuously refine categorization accuracy without model retraining.

Traffic Sign Robustness Study | Python, TensorFlow/Keras, OpenCV, scikit-learn | Jan 2025 – Apr 2025

- **Trained** and evaluated deep learning CNN and Spatial Transformer models on 9 image degradation scenarios to benchmark computer vision model robustness.

EDUCATION

Georgia State University | Master of Science in Computer Science (GPA: 4.13 / 4.00) | Atlanta, GA | Est Dec 2026

College of Engineering Roorkee | Bachelor of Technology in Computer Science | Roorkee, India

SKILLS & INTERESTS

Skills: C/C++, Python, Go, JavaScript, Shell Scripting, PowerShell, YAML, REST APIs, Microservices, Distributed Systems, CI/CD, GitLab, Docker, Kubernetes, AWS EC2, GCP, Azure, VMware vRealize, PostgreSQL, Oracle DB, MS SQL Server, ServiceNow, BMC TSSA, LDAP, RBAC, PyTest, ShellCheck, TensorFlow, OpenCV, Power BI